

QUESTIONS 22 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

INPUT

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9         System.out.println("Sum: " + a+b);
10        System.out.println("Difference: " + (a-b));
11        System.out.println("Product: " + a*b);
12        System.out.println("Quotient: " + a/b);
13    }
14 }
15 }
```

OUTPUT

```
Sum: 45
Difference: -1
Product: 20
Quotient: 0
```

4
5

QUESTIONS 22 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

```
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         System.out.println("enter the numbers: ");
9         int num=sc.nextInt();
10        int max=num;
11        for(int i=2;i<=n;i++)
12        {
13            num = sc.nextInt();
14            if(num>max)
15            {
16                max = num;
17            }
18        }
19        System.out.println("Maximum value: "+max);
20    }
21 }
```

OUTPUT

```
enter the numbers:
Maximum value: 18
Minimum value: 2
```

5
7
18
9
2
5

QUESTIONS 22 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

2004

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int year = sc.nextInt();
8         if(year % 400 == 0 || (year % 4==0 && year % 100 != 0))
9         {
10             System.out.println(year+" is Leap year");
11         }
12         else
13         {
14             System.out.println(year+" is not Leap Year");
15         }
16     }
17 }
```

OUTPUT

2004 is Leap year

QUESTIONS 22 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

INPUT

8

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int first=0;
9         int second =1;
10        System.out.println(first+" ");
11        System.out.println(second+" ");
12        for(int i=2;i<n;i++)
13        {
14            int third= first+second;
15            System.out.println(third);
16            first=second;
17            second=third;
18        }
19    }
20 }
```

OUTPUT

0
1
1
2
3
5
8
13

QUESTIONS 22 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int fact =1;
9         if(n==0 || n==1)
10        {
11            System.out.println("Factorial of "+n+" is:1");
12        }
13        for(int i=1;i<=n;i++)
14        {
15            fact=fact*i;
16        }
17        System.out.println("factorial of "+n+" is: "+fact);
18    }
19 }
```

OUTPUT

```
factorial of 5 is: 120
```

INPUT

5

QUESTIONS 22 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

```
**
***
****
*****
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int rows = sc.nextInt();
8         for(int i=1;i<=rows;i++)//outer loop
9         {
10            for(int j=1;j<=i;j++)//inner loop
11            {
12                System.out.print("**");
13            }
14            System.out.println();
15        }
16    }
17 }
```

OUTPUT

```
*
**
***
****
*****
```

INPUT

5

QUESTIONS 22 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9         int gcd=0;
10        for(int i=1;i<=a && i<=b;i++)
11        {
12            if(a%i==0 && b%i==0)
13            {
14                gcd = i;
15            }
16        }
17        System.out.println("GCD of "+a +" and "+b+" is "+gcd);
18    }
19 }
```

OUTPUT

```
GCD of 10 and 12 is 2
```

INPUT

```
10
12
```

QUESTIONS 22 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

5

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         int counter =0;
9         if(num==1)
10        {
11            System.out.println("neither prime nor composite");
12        }
13        for(int i=1;i<=num;i++)
14        {
15            if(num%i==0)
16            {
17                counter++;
18            }
19        }
```

```
20        if(counter==2)
21        {
22            System.out.println(num +" is prime number ");
23        }
24        else
25        {
26            System.out.println(num +" is composite number");
27        }
28    }
29 }
```

OUTPUT

5 is prime number

QUESTIONS 22 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         for(int i=1;i <=20;i++)
9         {
10             System.out.println(i + " X "+num+" = "+ num*i);
11         }
12     }
13 }
```

OUTPUT

```
10 X 1 = 10
10 X 2 = 20
10 X 3 = 30
10 X 4 = 40
10 X 5 = 50
10 X 6 = 60
10 X 7 = 70
10 X 8 = 80
10 X 9 = 90
10 X 10 = 100
```

INPUT

10

QUESTIONS 22 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int rows = sc.nextInt();
8         for(int i=rows;i>0;i--)
9         {
10             for(int j=1;j<=i;j++)
11             {
12                 System.out.print("*");
13             }
14             System.out.println();
15         }
16     }
17 }
```

OUTPUT

```
*****
****
***
**
*
```

INPUT

5

QUESTIONS 22 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int a = sc.nextInt();
8         int b = sc.nextInt();
9         int gcd=0;
10        for(int i=1;i<=a && i<=b;i++)
11        {
12            if(a%i==0 && b%i==0)
13            {
14                gcd=i;
15            }
16        }
17        int lcm = (a*b)/gcd;
18        System.out.println("LCM of "+a+" and "+b+" is "+lcm);
19    }
20 }
21 }
```

OUTPUT

LCM of 5 and 9 is 45

INPUT

5
9

QUESTIONS 22 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num=sc.nextInt();
8         int binary = 0;
9         int place = 1;
10        while(num>0)
11        {
12            binary=binary+(num%2)*place;
13            num=num/2;
14            place=place*10;
15        }
16        System.out.println("Binary = "+binary);
17    }
18 }
19 }
20 }
```

OUTPUT

Binary = 1111

INPUT

15

QUESTIONS 22 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num=sc.nextInt();
8         int decimal=0;
9         int base =1;
10        while(num>0)
11        {
12            decimal = decimal+(num % 10)*base;
13            num = num/10;
14            base = base*2;
15        }
16        System.out.println("Decimal: "+decimal);
17    }
18 }
```

OUTPUT

Decimal: 15

INPUT

1111

QUESTIONS 22 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         int sum =0;
9         while(num!=0)
10        {
11            sum=sum+num%10;
12            num=num/10;
13        }
14        System.out.println("Sum of Digits: "+sum);
15    }
16 }
```

OUTPUT

Sum of Digits: 6

INPUT

123

QUESTIONS 22 / 31 completed

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         int product=1;
9         while(num!=0)
10        {
11            product=product*(num%10);
12            num=num/10;
13        }
14        System.out.println("Product of digits: "+product);
15    }
16 }
```

OUTPUT

Product of digits: 6

INPUT

632

QUESTIONS 22 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         float radius = sc.nextFloat();
8         System.out.println("Area of circle: "+3.14*radius*radius );
9     }
10 }
```

OUTPUT

Area of circle: 78.5

INPUT

5

QUESTIONS 22 / 31 completed

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         float radius = sc.nextFloat();
8         float value = (float)4/3;
9         System.out.println("volume of Sphere: "+value*3.14*radius*radius*radius);
10    }
11 }
```

OUTPUT

```
volume of Sphere: 523.333348929882
```

INPUT

5

QUESTIONS 22 / 31 completed

RT-num.Pattern

RT- NUM.PATTERN

Write a Java program to print the following pattern using loops:

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int rows = sc.nextInt();
8         for(int i=1;i<=rows;i++)
9         {
10            for(int j=1;j<=i;j++)
11            {
12                System.out.print(i);
13            }
14            System.out.println();
15        }
16    }
17 }
```

OUTPUT

```
1
22
333
4444
55555
```

INPUT

5

QUESTIONS 22 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

10

```
1 import java.util.Scanner;  
2 public class main  
3 {  
4     public static void main(String[] args)  
5     {  
6         Scanner sc = new Scanner(System.in);  
7         int num=sc.nextInt();  
8         int sum=0;  
9         for(int i=1;i<num;i++)  
10        {  
11            if(num%i==0)  
12            {  
13                sum+=i;  
14            }  
15        }  
16        if(sum == num)  
17        {  
18            System.out.println(num+" is Perfect Number");  
19        }  
20        else  
21        {
```

```
20        else  
21        {  
22            System.out.println(num+" is Not a Perfect Number");  
23        }  
24    }  
25 }
```

OUTPUT

Perfect Number

QUESTIONS 22 / 31 completed

Palindrome ▾

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         int original=num;
9         int rev=0;
10        while(num!=0)
11        {
12            rev = rev*10+num%10;
13            num = num/10;
14        }
15        System.out.println("Reversed Number: "+rev);
16        if(original==rev)
17        {
18            System.out.println("Palindrome");
19        }
```

```
20        else
21        {
22            System.out.println("Not a Palindrome");
23        }
24    }
25 }
```

OUTPUT

```
Reversed Number: 321
Not a Palindrome
```

INPUT

12321

QUESTIONS 22 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

INPUT

1634

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         int original = num;
9         int num1 = num;
10        int sum=0;
11        int count=0;
12        while(num!=0)
13        {
14            count++;
15            num=num/10;
16        }
```

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
17        while(num1!=0)
18        {
19            int rem= num1%10;
20            sum+=Math.pow(rem,count);
21            num1=num1/10;
22        }
23        if(sum==original)
24        {
25            System.out.println("Armstrong number");
26        }
27        else
28        {
29            System.out.println("Not a Armstrong number");
30        }
31    }
32 }
33 }
```

OUTPUT

Not a Armstrong number

QUESTIONS 22 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int base=sc.nextInt();
8         int exponent = sc.nextInt();
9         int power=1;
10        System.out.println(base +" power "+exponent+"= "+Math.pow(base,exponent)); //inbuilt
11        //using loop
12        for(int i=1;i<=exponent;i++)
13        {
14            power = power*base;
15        }
16        System.out.print("power: "+power );
17    }
18 }
```

OUTPUT

```
2 power 0= 1.0
power: 1
```

INPUT

```
2
0
```

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class main{
3     static int fact(int n)
4     {
5         if(n==0||n==1)
6         {
7             return 1;
8         }
9         else
10        {
11            return n*fact(n-1);
12        }
13    }
14    public static void main(String[] args)
15    {
16        Scanner sc = new Scanner(System.in);
17        int n=sc.nextInt();
18        System.out.println("Factorial of "+n+" is "+fact(n));
19    }
20 }
21
```

OUTPUT

```
Factorial of 6 is 720
```

INPUT

```
6
```

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.*;
2 public class selection
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[]=new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        // selection sort - finding minimum value and swapping
14        for(int i=0;i<n-1;i++)
15        {
16            int min_idx=i;
```

INPUT

5
6
9
2
7
1

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
16            int min_idx=i;
17            for(int j=i+1;j<n;j++)
18            {
19                if(arr[j]<arr[min_idx])
20                {
21                    min_idx=j;
22                }
23            }
24            if(min_idx!=i)
25            {
26                int temp = arr[min_idx];
27                arr[min_idx]=arr[i];
28                arr[i] = temp;
29            }
30        }
31        for(int i=0;i<n;i++)
32        {
33            System.out.println(arr[i]);
34        }
35    }
36 }
```

OUTPUT

1
2
6
7
9

5
6
9
2
7
1

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();// number of elements
8         int arr[] = new int[n]; //array declaration
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        int max = arr[0];
14        for(int i=1;i<n;i++)
15        {
16            if(arr[i]>max)
17            {
18                max=arr[i];
19            }
20        }
21        System.out.println("Maximum: "+max);
22    }
23 }
```

OUTPUT

```
Maximum: 100
```

5
100
0
99
1
50

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        int min = arr[0];
14        for(int i=0;i<n;i++)
15        {
16            if(arr[i]<min)
17            {
18                min = arr[i];
19            }
20        }
21        System.out.println("Minimum: "+min);
22    }
23 }
```

OUTPUT

```
Minimum: 0
```

5
1
0
9
4
6

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        for(int i=0;i<n-1;i++)
14        {
15            for(int j=0;j<n-1-i;j++)
16            {
17                if(arr[j]>arr[j+1])
18                {
19                    int temp = arr[j];
20                    arr[j] = arr[j+1];
21                    arr[j+1] = temp;
```

```
21            arr[j+1] = temp;
22        }
23    }
24 }
25 System.out.println("Second Largest number: "+arr[n-2]);
26 }
27 }
```

OUTPUT

```
Second Largest number: 9
```

5
9
7
11
4
5

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr [] = new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        for(int i=0;i<n-1;i++)
14        {
15            for(int j=0;j<n-1-i;j++)
16            {
17                if(arr[j]>arr[j+1])
18                {
19                    int temp = arr[j];
20                    arr[j] = arr[j+1];
```

```
                arr[j+1] = temp;
19                }
20            }
21        }
22        System.out.println("Second Smallest element: "+arr[1]);
23    }
24 }
```

OUTPUT

```
Second Smallest element: 3
```

5
8
3
5
2
4

Sort-B



SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class sort
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[] = new int[n]; // declaration of array
9         for(int i=0; i<n; i++) // getting the array input values
10        {
11            arr[i] = sc.nextInt();
12        }
13        // bubble sort logic
14        for(int i=0; i<n-1; i++) // outer loop
15        {
16            for(int j=0; j<n-1-i; j++)
17            {
18                if(arr[j] > arr[j+1])
19                {
20                    int temp = arr[j];
```

```
20                    int temp = arr[j];
21                    arr[j] = arr[j+1];
22                    arr[j+1] = temp;
23                }
24            }
25        }
26        for(int i=0; i<n; i++)
27        {
28            System.out.println(arr[i]);
29        }
30    }
31 }
```

OUTPUT

```
9
7
5
3
2
```

```
5
5
7
3
9
2
```

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i] = sc.nextInt();
12        }
13        // insertion sort
14        for(int i=0;i<n;i++)
15        {
16            int key = arr[i];
17            int j=i-1;
18            while(j>=0 && arr[j]>key)
19            {
20                arr[j+1]=arr[j];
```

```
21                j=j-1;
22            }
23            arr[j+1] = key;
24        }
25        for(int i=0;i<n;i++)
26        {
27            System.out.println(arr[i]);
28        }
29    }
30 }
```

OUTPUT

```
2
4
5
7
8
```

```
5
2
7
5
4
8
```



SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.*;
2 public class main// for only sorted array
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int arr[] = new int[n];
9         for(int i=0;i<n;i++)
10        {
11            arr[i]=sc.nextInt();
12        }
13        int low =0;
14        int high =n-1;
15        int counter=0;
16        int target = sc.nextInt();//element need to search
17        while(low<=high)
18        {
19            int mid=(low+high)/2;
20            if(arr[mid]==target)
```

5
1
2
3
4
5
4

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
22         System.out.println("Element "+arr[mid]+" is found at index "+mid);
23         counter++;
24         break;
25     }
26     else if(arr[mid]<target)
27     {
28         low=mid+1;
29     }
30     else
31     {
32         high=mid-1;
33     }
34     }
35     if(counter==0)
36     {
37         System.out.println("element not found");
38     }
39 }
40 }
```

OUTPUT

```
Element 4 is found at index 3
```